

Geography Ch-20 (Advanced): India's Wheat Granary - The Indo-Gangetic Plain & Punjab-Haryana Belt

Tier 2: Advanced | Source: D.R. Khullar + Majid Husain (Geography of India)
India-applied grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

India's 'Steppe Analogue': The Wheat Granary of Punjab-Haryana & the Indo-Gangetic Plain

GS-I (Geography of India) + GS-III (Agriculture) Geography

■ NEWS TRIGGER

Advanced application (D.R. Khullar; Majid Husain). India has no true temperate-steppe climate, but the same ECONOMIC role - the nation's wheat granary - is played by the alluvial Indo-Gangetic plain, above all the Punjab-Haryana belt, aided by Green Revolution technology and irrigation.

■ INTRO & ORIGIN

The steppe chapter's key idea is the 'granary of the world'. India reproduces this as its own granary - not on chernozem grassland but on the fertile alluvium of the Sutlej-Ganga plains, where winter (Rabi) wheat under assured irrigation makes Punjab-Haryana the country's breadbasket.

India's wheat is a RABI (winter) crop needing cool growing weather and warm ripening - climatically similar to steppe wheat. The Green Revolution (HYV seeds, fertiliser, tube-well + canal irrigation, MSP/procurement) turned the NW plains into a surplus granary.

■ KEY DATA

Element	Detail	Note
Core belt	Punjab, Haryana, W. Uttar Pradesh	India's wheat granary
Other producers	Madhya Pradesh, Rajasthan, Bihar	Rising shares
Crop season	Rabi (winter): sown Oct-Dec, harvest Mar-Apr	Cool growth, warm ripening
Enablers	HYV seeds, irrigation, fertiliser, MSP	Green Revolution package
Soil	Alluvial (not chernozem)	Renewed by Himalayan rivers

■ STATIC THEORY

- **Wheat as a Rabi crop:** needs a cool, moist growing season and a warm, dry ripening period - which is why it is grown in winter in NW India, echoing the temperate wheat regime.
- **The granary belt:** Punjab, Haryana and western UP dominate output and, crucially, procurement; Madhya Pradesh and Rajasthan are rising. This is India's functional equivalent of the prairie/steppe wheat belt.
- **Green Revolution:** HYV (semi-dwarf) wheat (Borlaug/derived varieties), chemical fertiliser, assured canal + tube-well irrigation, and price support (MSP) transformed the NW plains into a surplus granary in the late 1960s-70s.
- **Soil contrast:** unlike the steppe's chernozem, India's granary sits on ALLUVIAL soil constantly renewed by Himalayan rivers - fertile but demanding heavy irrigation.
- **Sustainability strain:** the Punjab-Haryana wheat-rice system now faces groundwater depletion, soil-nutrient imbalance and stubble-burning - the ecological cost of granary intensification.

■ MUST-KNOW FACTS

1. India's wheat granary = Punjab, Haryana, western UP (Indo-Gangetic alluvial plain).
2. Wheat is a RABI (winter) crop; sown Oct-Dec, harvested Mar-Apr.

■ UPSC TRAPS

- ✗ India's wheat belt sits on black chernozem soil like the steppe.
- ✓ It sits on ALLUVIAL soil of the Indo-Gangetic plain; India's black (regur) soil is in the Deccan and grows cotton, not the wheat belt.

3. Green Revolution (HYV + irrigation + fertiliser + MSP) created the surplus granary.

4. Soil is ALLUVIAL (river-renewed), not steppe chernozem.

5. Intensive wheat-rice cropping causes groundwater depletion & stubble-burning in Punjab-Haryana.

✗ Wheat in India is a Kharif (monsoon) crop.

✓ Wheat is a RABI (winter) crop; rice is the Kharif crop in the same belt.

■ MAINS ANGLE

Explain how the Indo-Gangetic plain became India's wheat granary despite lacking a steppe climate, and assess the sustainability crisis of the Punjab-Haryana wheat-rice system.

■ STUDY LINK

Geography -> Indian Agriculture -> Wheat / Green Revolution belt; India's analogue of the world's granaries

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: India's Record 2025-26 Wheat Procurement Despite Heat Stress

GS-III
(Agriculture / Food Security) +
GS-I (Geography of India)
Geography -
Current Affairs

■ NEWS TRIGGER

2025-26: despite pre-harvest heat, unseasonal rain and hail, India's wheat procurement hit a multi-year high (~35 million tonnes, up ~17% YoY) against a raised 34.5 mt target; MSP was Rs 2,425/quintal; ~67% was bought under RELAXED quality norms due to weather damage; buffer stocks stayed well above norms (wheat ~358 lakh mt vs ~275.8 lakh mt norm on 1 July).

■ INTRO & ORIGIN

India's 2025-26 wheat season is the living application of the granary chapter: it shows how a breadbasket copes with climate shocks through policy - MSP, raised procurement targets and relaxed norms - to keep buffer stocks and food security intact.

Wheat output was hit by heatwaves (grain shrivelling) - a climate risk to the granary. Government raised the procurement target, relaxed quality norms and used MSP to protect farmers and buffer stocks.

■ KEY DATA

Metric	2025-26 value	Note
Procurement	~35 mt (+~17% YoY)	Exceeded 34.5 mt target
MSP (wheat)	Rs 2,425 / quintal	Some states add bonus
Relaxed-norm share	~67% of procurement	Weather-damaged grain
Buffer stock (1 Jul)	~358 lakh mt vs ~275.8 norm	Above buffer norm

■ STATIC THEORY

- **MSP & procurement:** the government buys wheat at a Minimum Support Price through FCI/state agencies to assure farmer income and build the central pool - the policy backbone of India's granary.
- **Buffer stock norms:** minimum wheat/rice the central pool must hold (for PDS + food security); 2025-26 stocks stayed comfortably above norm despite the heat-hit crop.
- **Climate risk to the granary:** terminal-heat stress during grain-filling shrivels wheat and cuts yield/quality - forcing relaxed procurement norms; the direct climate-agriculture link.

■ MUST-KNOW FACTS

1. 2025-26 wheat procurement ~35 mt (up ~17% YoY), exceeding the 34.5 mt target.
2. Wheat MSP 2025-26 = Rs 2,425/quintal (states may add bonus).
3. ~67% procured under RELAXED quality norms due to heat/rain damage.
4. Buffer stock (1 Jul) ~358 lakh mt vs ~275.8 lakh mt norm - well above.
5. Punjab, Haryana, MP, Rajasthan, UP were the main procuring states.

■ UPSC TRAPS

- ✗ Heat damage caused India's wheat buffer to fall below norms in 2025-26.
- ✓ Despite heat losses, coordinated procurement kept buffer stocks WELL ABOVE norms (~358 vs ~275.8 lakh mt).
- ✗ MSP is the market price of wheat.
- ✓ MSP is a government-assured FLOOR price for procurement, not the open-market price.

■ MAINS ANGLE

Discuss how MSP, procurement and buffer-stocking insulate India's wheat granary from climate shocks, using the heat-stressed 2025-26 season as a case study.

■ STUDY LINK

Geography -> Indian wheat granary + GS-III food security (MSP, FCI, buffer stocks); climate-heat link to Ch-13 CA

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